

Aniruddh Ballal

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Engineer working across physics and production software, from solar coronal-field simulations and orbital mechanics to full-stack financial systems and modelling. Strong in end-to-end development, scientific simulation, and debugging complex systems, with a focus on understanding how they behave.

Education

Manipal Institute of Technology — Manipal, KA
B.Tech, Information Technology — *Expected Jul 2026*
CGPA: 8.1

Deeksha Integrated PU College — Yelahanka, KA
Karnataka 2nd PUC (94.2%) — 2022

Daffodils English School — Sanjaynagar, KA
ICSE 10th (92.9%) — 2020

Skills

Languages

TypeScript, Python, JavaScript, Rust, Kotlin, C/C++, SQL

Frontend & Mobile

React, Next.js, React Native, Three.js, Tailwind CSS, MUI, Redux Toolkit

Backend & Data

FastAPI, Node.js, Express, PostgreSQL, SQLAlchemy, MongoDB, Firebase

Scientific & Systems

NumPy, SciPy, XGBoost, PuLP, Astropy, Matplotlib, WASM, Web Workers, GLSL, Git, Docker, KiCad

Experience

Assistant Software Engineer | Indiassetz Mar 2026–Oct 2026
Modernised React Native applications and extended an existing CRM with payment reviews, approvals and revenue reporting. Developed a full-stack investor/back-office platform covering document governance, capital-call settlement tracking and portfolio reporting.

Data Analyst Intern | Qwixpert Consulting Jan–Mar 2026
Analysed client data, automated reporting across Excel workbooks and prepared consulting presentations. Independently built decision-support tools for supply-chain network design, demand forecasting and dispatch planning.

Frontend Web Dev Intern | Indiassetz Jun–Jul 2025
Collaborated with design and engineering teams to enhance UI/UX with React and Material UI; implemented micro-interactions and component redesigns using clean, reusable, version-controlled code.

Solar Physics Intern | Dept. of Physics, IISc Jun–Aug 2024
Worked under Dr. Prantika Bhowmik on a Python project modelling and visualising solar magnetic-field evolution and cycles with spherical harmonics, SciPy, AstroPy, NumPy, and Matplotlib; derived quantitative insights into solar-cycle behaviour.

Community

Astronomy Club, MIT Manipal Feb 2024 – Apr 2025
Managing Committee Member; completed coursework in AstroTech, black holes, astrobiology, and data-driven astronomy; organised club events and outreach.

Revels PnP, MIT Manipal Jan – Mar 2024
Led a 4-person team to paint 10+ large acrylic posters and build 3D models for MIT's annual cultural festival (25,000+ attendees).

Manipal Runners' Club Dec 2023 – Feb 2024
Co-organised the Manipal Marathon 2024 (8,000+ participants) and coached enrollees through a 6-week training programme.

Junior Avionics Member | ThrustMIT Nov 2022–Nov 2023
Built avionics for student rocketry: custom PCBs, telemetry, RF/GPS/IMU integration, microcontrollers, and control theory.

Projects *demos & source: aniruddhballalportfolio.vercel.app*

Sunalyse 2025–26
Reconstructed coronal fields across 181 Carrington rotations with a Python PFSS pipeline and Three.js; vectorisation and four-core tracing achieved a documented **7.6x speedup** over the original implementation.

Queen's Gambit 2.0 2025–26
Built browser-based file steganography with legal chess moves, Rust/WASM and Web Workers; optional AES-256-GCM protects file contents. Documented 32.1 KB encoding benchmark: **195 s to 5.2 s**.

SatLies 2026
Built a Three.js orbital simulator with RK4 integration, comparing Earth's oblateness (J2) and lunar/solar gravity against a two-body baseline through live trajectories and separation plots.

Credex 2026
Built a Next.js/Node.js trade-finance prototype linking procurement, payment records and loan review; a Python/FastAPI service uses PuLP to schedule payments under cash-flow and deadline constraints.

kNupdate 2026
Built a Kotlin/Compose activity-sharing app with paired-device delivery and SQLite histories; a Cloudflare relay discards payloads after all recipients acknowledge or 30 seconds elapse.

dedgram 2026
Built an Instagram research toolkit with Android capture, Playwright scraping and a public atlas; stored profile/reel snapshots, source evidence and username history to investigate suspected automated accounts.